Bluestock Mutual Fund Analytics Capstone

Comprehensive End-to-End Data Engineering & Analytics Pipeline

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Project: Bluestock Analytics

1. Executive Summary

This report documents the architectural design, engineering implementation, and advanced analytical findings of the Bluestock Mutual Fund Analytics Capstone Project.

The core objective was to aggregate highly fragmented mutual fund data representing over INR 81 Lakh Crore of Assets Under Management (AUM) into a centralized, analytical Star Schema. By achieving this, the platform provides institutional-grade risk metrics, interactive dashboarding, and automated reporting capabilities designed to drive data-driven investment decisions.

2. Data Engineering & ETL Pipeline

The data engineering pipeline was fully automated using Python, Pandas, and SQLite, ensuring robust data integrity:

- Data Ingestion: Automated extraction of 10 disparate CSV files encompassing historical NAVs, AUM figures, benchmarks, investor demographics, and transactions.
- Data Cleansing: Missing dates in NAV histories were programmatically forward-filled to handle non-trading days. Deduplication and outlier removal were strictly enforced.
- Dimensional Modeling: Data was meticulously mapped into a highly optimized 5-table Star Schema (fact_nav, fact_transactions, fact_performance, dim_fund, dim_date) to allow complex aggregations without performance bottlenecks.

3. Quantitative Financial Modeling

Extensive quantitative analysis was conducted to rank and evaluate the 40 mutual fund schemes:

- Performance Metrics: Calculated 1-year, 3-year, and 5-year Compounded Annual Growth Rates (CAGR) utilizing exact 252-day trading year formulas.
- Risk-Adjusted Returns: Computed annualized Sharpe and Sortino ratios against a 6.5% risk-free rate proxy to isolate true management outperformance.

- Benchmark Regressions: Executed Ordinary Least Squares (OLS) regressions against the NIFTY 100 benchmark to extract portfolio Alpha (excess returns) and Beta (market volatility correlation).
- Downside Protection: Analyzed historical Maximum Drawdowns (peak-to-trough drops) to evaluate fund resilience during macroeconomic shocks.
- Comprehensive Scorecard: Developed a proprietary 0-100 composite ranking system weighting trailing returns (30%), Sharpe ratios (25%), Alpha (20%), Expense Ratios (15%), and Drawdowns (10%).

4. Advanced Analytics & Cohort Modeling

To provide deeper insights beyond standard historical trailing returns, advanced analytics and stochastic models were implemented:

- Tail Risk Analysis (VaR): Calculated historical 95% Value-at-Risk (VaR) and Conditional VaR to quantify expected shortfall during extreme market downturns, exposing heavy tail-risks primarily in sectoral funds.
- Sector Concentration: Computed the Herfindahl-Hirschman Index (HHI) for fund portfolios to mathematically highlight concentration risk.
- Cohort Modeling: Grouped investors by their initial transaction year to monitor Systematic Investment Plan (SIP) stickiness, average step-up behaviors, and long-term brand loyalty.
- Early Churn Detection: Flagged 'At-Risk' investors who exhibited SIP payment gaps exceeding 35 days, providing actionable intelligence for customer retention teams.

5. Bonus Challenges Accomplished

Going above and beyond the baseline requirements, the following five bonus challenges were executed:

1. Cron Automation (B1): Scheduled automated pipeline scripts to fetch and append daily NAV metrics.
2. Interactive Web Application (B2): Deployed a real-time, 4-page Streamlit dashboard featuring interactive Plotly visualizations, dual-axis charts, and cross-filtering slicers.
3. Stochastic Projections (B3): Engineered a Monte Carlo simulation running 100 iterations of 5-year future NAV trajectories, generating 5th, 50th, and 95th percentile expected bounds.
4. Modern Portfolio Theory (B4): Programmed a Markowitz Efficient Frontier optimizer utilizing covariance matrices to discover the mathematical maximum Sharpe ratio portfolio allocation.
5. Automated Email Reporting (B5): Built a script to construct and send dynamic HTML email reports summarizing weekly performance directly to management.